

instrumentation tiers

**T1 boundary
patching**
wrap proxies

**T2 ufunc
proxy**
call/reduce/...

**T3 tracer
arrays**
taint, opt-in

**T4 runtime
monitor**
sys.monitoring

**T5 native
census**
LD_PRELOAD

**event
stream**
versioned

**append-only
capture**
JSONL, crash-safe

Parquet
per-run finalize

Zarr / npy
array payloads

**alignment +
attribution**
sig bits, amplif.

report / dashboard
check / diff (CI)
Perfetto export

one independent subprocess per repetition;
runs aligned call-by-call across repetitions

